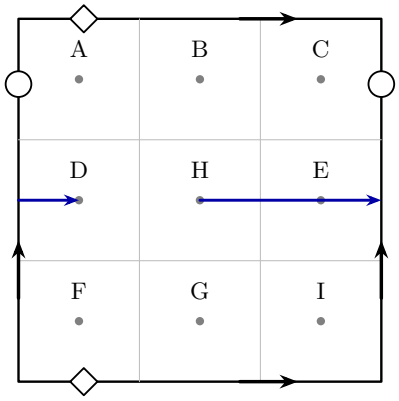
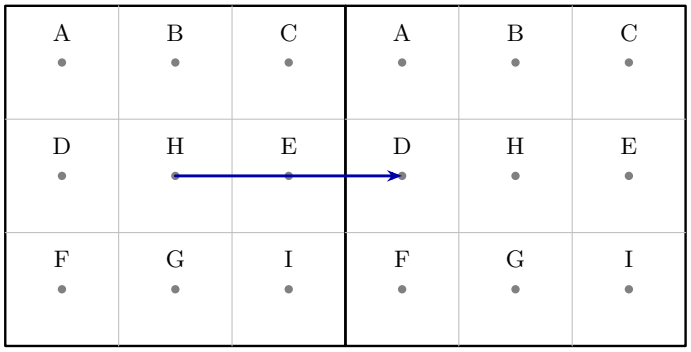
One arrow moves to the center of the next small square. Right meets left at the same height; top meets bottom at the same place across. Keep going the same way. Repeated copies continue beyond the page.

portal map



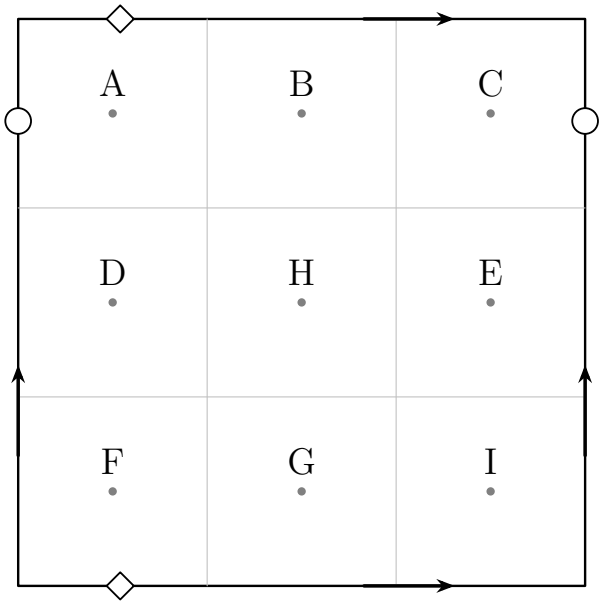
two steps right

repeated map



original copy next copy

Problem 1: Start at H for each trip. Predict where it ends, then try it with your pawn.



$\rightarrow \rightarrow \downarrow$ _____

$\uparrow \uparrow$ _____

$\leftarrow \downarrow$ _____

$\uparrow \uparrow \rightarrow \rightarrow$ _____

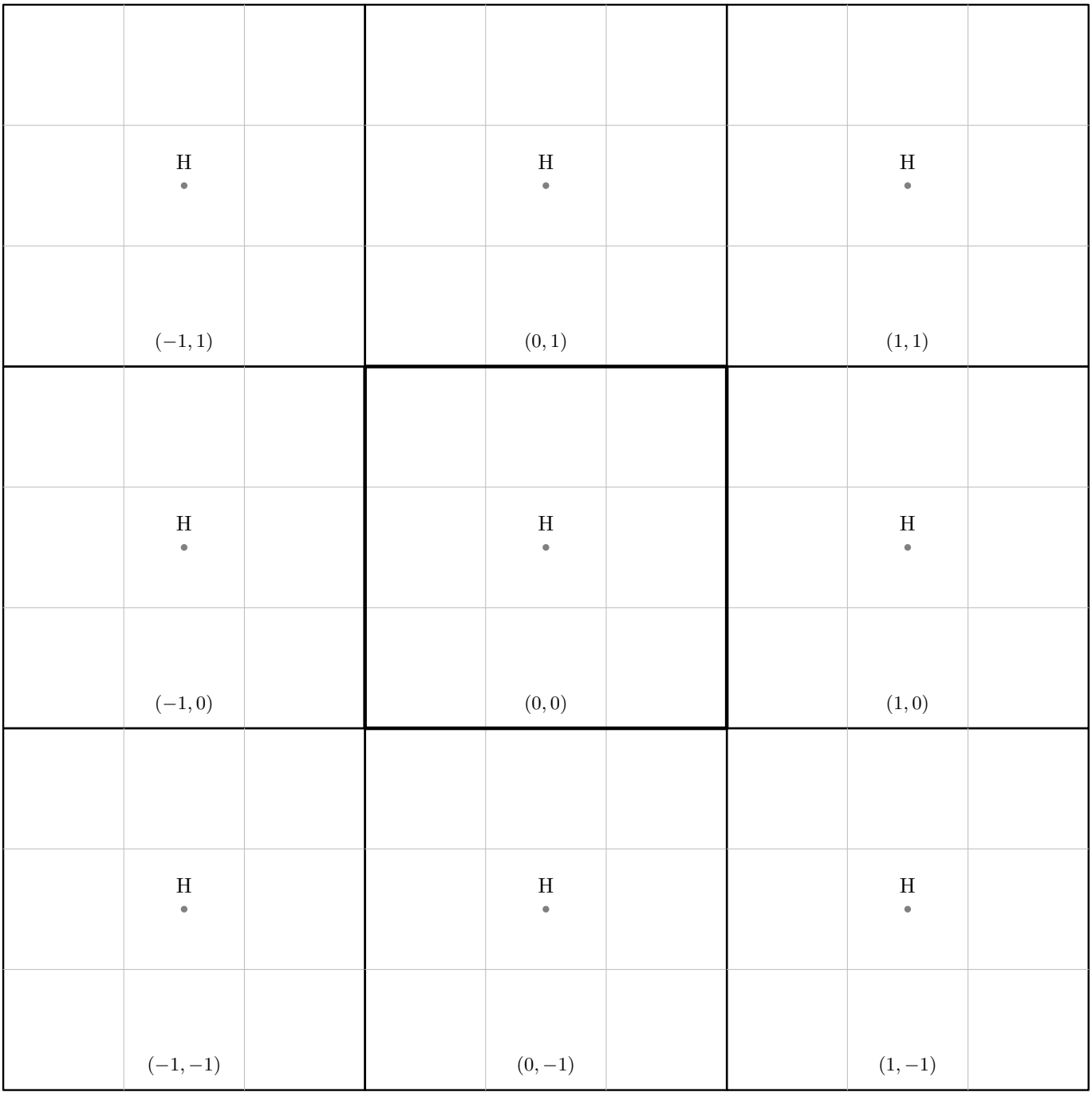
Problem 2: Start at H in copy (0,0). Find the finishing copy for each trip.

$\rightarrow \rightarrow \rightarrow \uparrow \uparrow \uparrow$

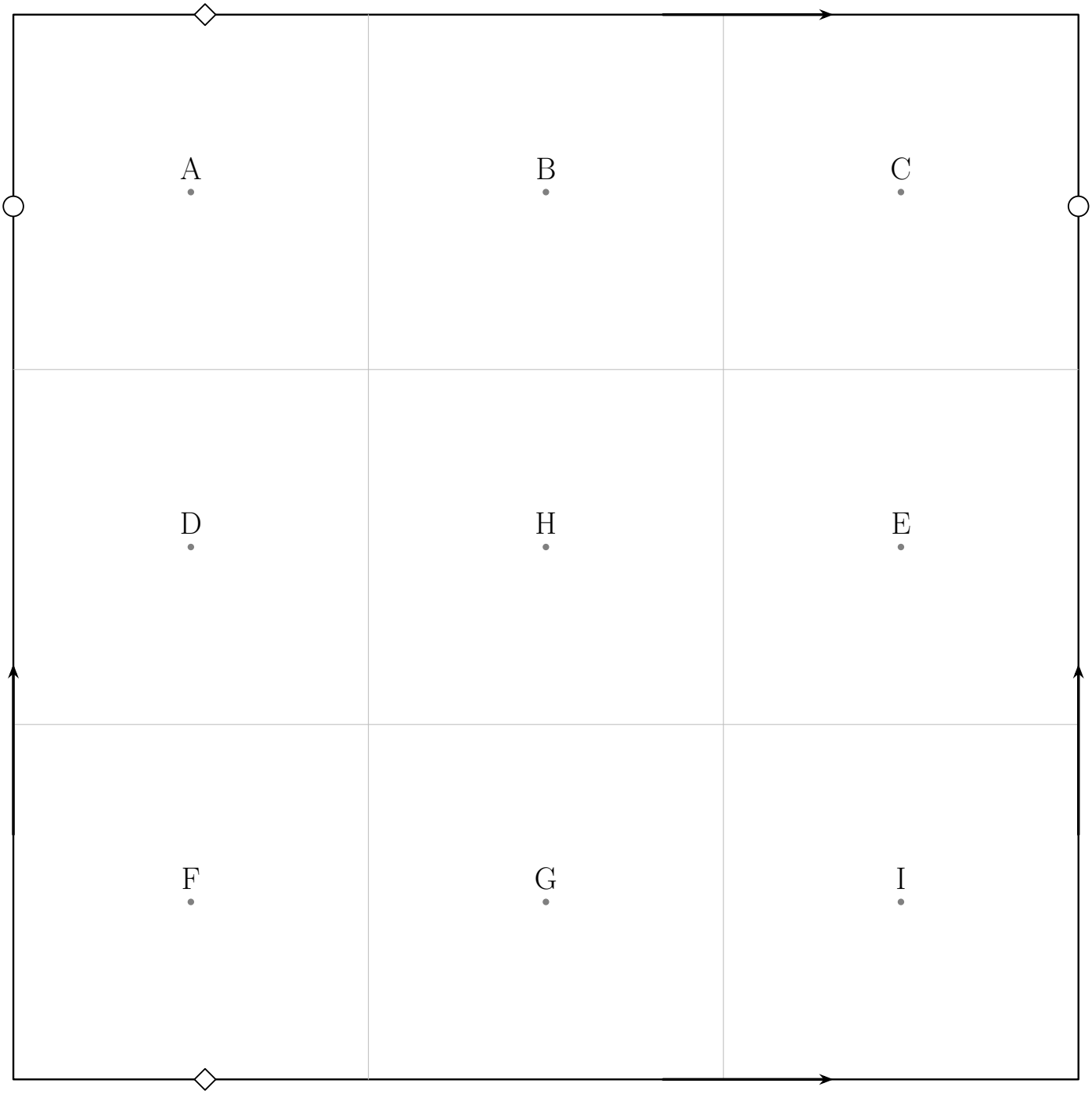
$\rightarrow \rightarrow \leftarrow \leftarrow$

$\uparrow \uparrow \uparrow \rightarrow \rightarrow \rightarrow$

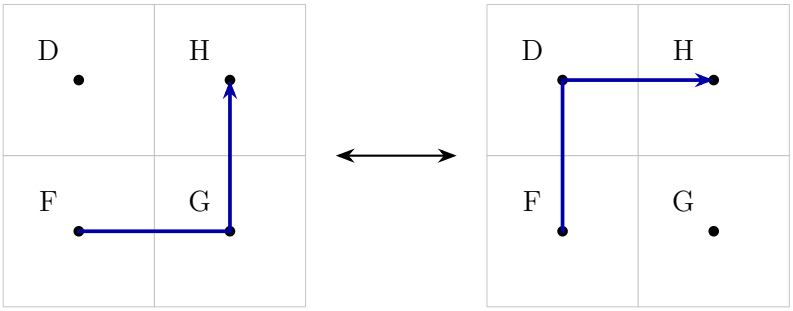
$\rightarrow \rightarrow \rightarrow \uparrow \uparrow \uparrow \leftarrow \leftarrow \leftarrow \downarrow \downarrow \downarrow$



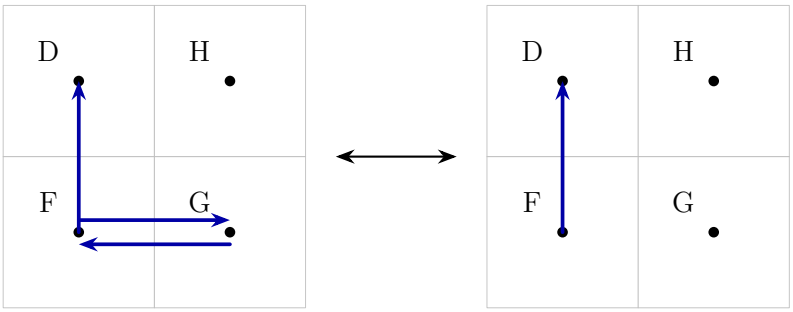
Problem 3: Make four different trips from H back to H. Choose trips whose replays finish in different copies of H on the repeated map.



Trips may move through the squares and cross themselves. Keep the start and finish of the whole trip at H, and keep its starting copy fixed. Two changes are allowed, either way:



Slide: swap two sides of one small square for its other two sides, in any rotation.



Backtrack: add or erase a step followed straight away by the opposite step.

Problem 4: Use tracing paper on the repeated map from page 2. Which trip can shrink all the way to staying at H? Show changes to the whole trip.

$\rightarrow \rightarrow \rightarrow \uparrow \uparrow \uparrow \leftarrow \leftarrow \leftarrow \downarrow \downarrow \downarrow$

$\rightarrow \rightarrow \rightarrow$

Problem 5: Change the first trip into the second using the allowed moves. Keep the same starting copy.

$\rightarrow \rightarrow \rightarrow \uparrow \uparrow \uparrow$ $\uparrow \uparrow \uparrow \rightarrow \rightarrow \rightarrow$

Problem 6: When can two trips from H back to H be changed into each other? Explain. Add copies if your trip leaves the page.

